

SIDDHANT RAI

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SUMMARY

Aspiring Machine Learning Engineer with practical experience building intelligent systems and deploying real-world applications. Created **Flashkick**, a football highlight generator using 3D CNNs, Whisper, and MoviePy. Built a **Movie Recommender System** using content-based filtering and deployed with Streamlit. Currently working on a **Book Recommender System** with collaborative filtering. Passionate about applying ML to solve real-world problems with clean, scalable code.

EDUCATION

Lalitpur Engineering College

Bachelor in Computer Engineering

2021–2025

Lalitpur, Nepal

SKILLS

Languages: Python, JavaScript, C++

Machine Learning Libraries: scikit-learn, pandas, NumPy, Seaborn, Matplotlib

Deep Learning: TensorFlow, Keras, CNNs, LSTM, GRU, RNNs, Embedding Layers, Sequence Modeling

Natural Language Processing: OpenAI Whisper, NLTK, CountVectorizer, TF-IDF

Recommender Systems: Content-Based Filtering, Collaborative Filtering, Matrix Factorization, Cosine Similarity

Web Frameworks: Flask, Streamlit

Tools and Platforms: Git, MoviePy, Google Colab, Heroku, Netlify, Kaggle

PROJECTS

Flashkick Football Highlight Generator — *React.js, Flask, 3D CNN*

Completed

- Built a 3D CNN model on football videos achieving 92% key-moment detection accuracy.
- Lowered the highlight generation time by 45% using optimized frame sampling.

Movie Recommender System — *Content-Based Filtering, Streamlit*

Completed

- Built a content-based movie recommender using Kaggle MovieLens (100k ratings, 1,682 movies).
- Utilized CountVectorizer and cosine similarity to generate personalized recommendations.
- Deployed an interactive Streamlit app for user evaluation.

Book Recommender System — *Collaborative Filtering, Cosine Similarity*

Completed

- Built a book recommender using Kaggle Goodbooks-10k (10k books, 6M ratings from 53k users).
- Employed collaborative filtering with cosine similarity, improving recommendation relevance by 25%.
- Prepared a user-item matrix and lessened sparsity for better predictions.

Next-Word Prediction using LSTM — *TensorFlow, Keras, Quotes Dataset*

Completed

- Built an LSTM model to predict the next word in sentences using Kaggle Quotes dataset (20k quotes).
- Put into practice tokenization, padding, and sequence generation; achieved 35% top-1 and 60% top-3 accuracy.
- Designed LSTM with embedding layers, 2 LSTM layers, and dense softmax output.
- Deployed a web demo for real-time next-word prediction.

Vendor Purchase and Sales Data Analysis — *SQL, Python*

Completed

- Analyzed vendor performance using SQL and Python (Pandas).
- Generated data summaries and visualizations to highlight trends and key metrics.
- Identified opportunities to optimize procurement, projecting 15% cost reduction.

WORKSHOPS & TRAININGS

90-day Free Course – AI/ML (2025)

- Implemented end-to-end ML pipelines on small datasets, improving workflow efficiency by 20%.
- Applied predictive and recommender systems using realistic datasets, achieving 75% recommendation accuracy.
- Refined model evaluation and tuning processes, increasing performance on test sets by 10%.
- Deployed interactive ML applications using Streamlit, enabling 15 user interactions per session during testing.

45-day Free Course – Web Development (2025)

- Built 4 projects using JavaScript and React, increasing code reusability and reducing bugs by 15%.
- Used state management with React hooks and custom hooks to modularize components.
- Refined DOM manipulation, event handling, and CRUD operations to streamline UI updates.
- Implemented data persistence with localStorage, reducing session reload issues by 10%.

MERN Training – Lalitpur Engineering College (2022)

- Built full-stack apps with MongoDB, Express, React, and Node.js, completing 1 project.
- Developed REST APIs and integrated frontend-backend for 3 modules, evaluated with 3 users.
- Gained experience with Git, improving version control practices during team exercises.

AWARDS & CERTIFICATIONS

- **Certified AWS Cloud Foundation**
- **React.js Certificate** — Complete Coding
- **JavaScript Certificate** — Complete Coding
- **Certified AWS Cloud Practitioner from BroadwayInfosys**

PARTICIPATION

- **Poster Presentation at “International Conference on Engineering Technology & Infrastructure Development (2024)”**